Force Torque Thr Force Mapping

Component Design Document

1 Description

Force torque thruster force mapping converts a commanded body force and torque into per-thruster force commands using the truncated-SVD pseudo-inverse of the thruster control mapping matrix.

2 Requirements

No requirements have been specified for this component.

3 Design

3.1 At a Glance

Below is a list of useful parameters and statistics that give a quick look into the makeup of the component.

- **Execution** - *passive*
- **Number of Connectors** - 5
- **Number of Invokee Connectors** - 3
- **Number of Invoker Connectors** - 2
- **Number of Generic Connectors** - *None*
- **Number of Generic Types** - *None*
- **Number of Unconstrained Arrayed Connectors** - *None*
- **Number of Commands** - *None*
- **Number of Parameters** - 6
- **Number of Events** - *None*
- **Number of Faults** - *None*
- **Number of Data Products** - 1
- **Number of Data Dependencies** - 2
- **Number of Packets** - *None*

3.2 Diagram



Figure 1: Force Torque Thr Force Mapping component diagram.

3.3 Connectors

Below are tables listing the component's connectors.

3.3.1 Invokee Connectors

The following is a list of the component's *invokee* connectors:

Table 1: Force Torque Thr Force Mapping Invokee Connectors

Name	Kind	Type	Return_Type	Count
Tick_T_Recv_Sync	recv_sync	Tick.T	-	1
Reset_Tick_T_ Recv_Sync	recv_sync	Tick.T	-	1
Parameter_ Update_T_Modify	modify	Parameter_ Update.T	-	1

Connector Descriptions:

- **Tick_T_Recv_Sync** - Run the algorithm up to the current time.
- **Reset_Tick_T_Recv_Sync** - Apply any pending parameter update. Called on GNC state change.
- **Parameter_Update_T_Modify** - The parameter update connector.

3.3.2 Invoker Connectors

The following is a list of the component's *invoker* connectors:

Table 2: Force Torque Thr Force Mapping Invoker Connectors

Name	Kind	Type	Return_Type	Count
Data_Product_ Fetch_T_Request	request	Data_Product_ Fetch.T	Data_Product_ Return.T	1
Data_Product_T_ Send	send	Data_Product.T	-	1

Connector Descriptions:

- **Data_Product_Fetch_T_Request** - Fetch a data product item from the database.
- **Data_Product_T_Send** - The data product invoker connector

3.4 Interrupts

This component contains no interrupts.

3.5 Initialization

Below are details on how the component should be initialized in an assembly.

3.5.1 Component Instantiation

This component contains no instantiation parameters in its discriminant.

3.5.2 Component Base Initialization

This component contains no base class initialization, meaning there is no `init_Base` subprogram for this component.

3.5.3 Component Set ID Bases

This component contains commands, events, packets, faults, or data products that require a base identifier to be set at initialization. The `set_Id_Bases` procedure must be called with the following parameters:

Table 3: Force Torque Thr Force Mapping Set Id Bases Parameters

Name	Type
`Data_Product_Id_Base`	`Data_Product_Types.Data_Product_Id_Base`
`Parameter_Id_Base`	`Parameter_Types.Parameter_Id_Base`

Parameter Descriptions:

- **`Data_Product_Id_Base`** - The value at which the component's data product identifiers begin.
- **`Parameter_Id_Base`** - The value at which the component's parameter identifiers begin.

3.5.4 Component Map Data Dependencies

This component contains data dependencies that need their IDs mapped to valid data product IDs at initialization. Each data dependency also needs a stale limit provided in microseconds. If the timestamp of the data dependency is older than the reference time (usually the current time) minus the stale limit then the data dependency is considered too old to safely use. The `map_Data_Dependencies` procedure must be called with the following parameters:

Table 4: Force Torque Thr Force Mapping Map Data Dependencies Parameters

Name	Type
`Commanded_Torque_Id`	`Data_Product_Types.Data_Product_Id`
`Commanded_Torque_Stale_Limit`	`Ada.Real_Time.Time_Span`
`Commanded_Force_Id`	`Data_Product_Types.Data_Product_Id`
`Commanded_Force_Stale_Limit`	`Ada.Real_Time.Time_Span`

Parameter Descriptions:

- **`Commanded_Torque_Id`** - The external data product ID used to fetch the value for the 'Commanded_Torque' data dependency.
- **`Commanded_Torque_Stale_Limit`** - The stale limit used for the 'Commanded_Torque' data dependency. If the fetched data product is older than the given Time_Span then it is deemed stale and unfit for use by the component.
- **`Commanded_Force_Id`** - The external data product ID used to fetch the value for the 'Commanded_Force' data dependency.

- **Commanded_Force_Stale_Limit** - The stale limit used for the 'Commanded_Force' data dependency. If the fetched data product is older than the given Time_Span then it is deemed stale and unfit for use by the component.

3.5.5 Component Implementation Initialization

The calling of this implementation class initialization procedure is mandatory. Initializes the force torque thruster force mapping algorithm. The `init` subprogram takes no parameters.

3.6 Commands

The Force Torque Thr Force Mapping component has no commands.

3.7 Parameters

Parameters for the Force Torque Thr Force Mapping component

Table 5: Force Torque Thr Force Mapping Parameters

Local ID	Parameter Name	Type	Default Value
0x0000 (0)	Num_Thrusters	Packed_U32.T	(Value=>8)
0x0001 (1)	R_Thruster_B	Packed_F32x24.T	[-1.0,-1.0,+1.0,-1.0,-1.0,+1.0,+1.0,+1.0,+1.0,+1.0,+1.0,+1.0,+1.0,-1.0,+1.0,+1.0,-1.0,-1.0,-1.0,-1.0,-1.0,-1.0]
0x0002 (2)	T_Hat_Thruster_B	Packed_F32x24.T	[+0.0,+1.0,+0.0,+0.0,+0.0,-1.0,+0.0,+0.0,-1.0,-1.0,+0.0,+0.0,+0.0,-1.0,+0.0,-1.0,+0.0,+0.0,+1.0,+0.0,+0.0,+0.0,+1.0,+0.0]
0x0003 (3)	Center_Of_Mass_B	Packed_F32x3.T	[0.0,0.0,0.0]
0x0004 (4)	Desired_Control_Axes_B	Desired_Control_Axes.T	(Torque_X=>True, Torque_Y=>True, Torque_Z=>True, Force_X=>True, Force_Y=>True, Force_Z=>True)
0x0005 (5)	Thruster_Availability	Thruster_Availability_X8.T	[others=>Force_Torque_Thr_Force_Mapping_Enums.Device_Availability.Available]

Parameter Descriptions:

- **Num_Thrusters** - [-] Number of configured thrusters. Must be in [1, 8]. Only the first Num_Thrusters entries of the geometry take part in the mapping.

- **R_Thruster_B** - [m] Per-thruster locations in the body frame, 3 components per thruster in row major order.
- **T_Hat_Thruster_B** - [-] Per-thruster thrust directions in the body frame, 3 components per thruster in row major order. Each row must be a unit vector to within 1e-3.
- **Center_Of_Mass_B** - [m] Center of mass in the body frame. Must be finite.
- **Desired_Control_Axes_B** - [-] Per-axis controllability assertions. An asserted axis must lie in the column space of the thruster control mapping matrix.
- **Thruster_Availability** - [-] Per-thruster availability. An unavailable thruster is left out of the mapping and always receives a zero force command. A minimum of one thruster must be available.

3.8 Events

The Force Torque Thr Force Mapping component has no events.

3.9 Data Products

Data products for the Force Torque Thr Force Mapping component.

Table 6: Force Torque Thr Force Mapping Data Products

Local ID	Data Product Name	Type
0x0000 (0)	Thruster_Force_Cmd	Thr_Force_Cmd.T

Data Product Descriptions:
- **Thruster_Force_Cmd** - Thruster force command (8 thrusters)

3.10 Data Dependencies

Data dependencies for the Force Torque Thr Force Mapping component.

Table 7: Force Torque Thr Force Mapping Data Dependencies

Data Dependency Name	Type
Commanded_Torque	Cmd_Torque_Body.T
Commanded_Force	Cmd_Force_Body.T

Data Dependency Descriptions:
- **Commanded_Torque** - Commanded control torque in body frame components
- **Commanded_Force** - Commanded control force in body frame components

3.11 Packets

The Force Torque Thr Force Mapping component has no packets.

3.12 Faults

The Force Torque Thr Force Mapping component has no faults.

4 Unit Tests

The following section describes the unit test suites written to test the component.

4.1 *Force_ Torque_ Thr_ Force_ Mapping_ Tests* Test Suite

This is a unit test suite for the Force Torque Thr Force Mapping component

Test Descriptions:

- **`Test_Pure_Torque`** - A commanded torque about each body axis is reproduced by the mapped thruster forces.
- **`Test_Pure_Force`** - A commanded force along each body axis is reproduced by the mapped thruster forces.
- **`Test_Combined_Force_And_Torque`** - A simultaneous force and torque command is reproduced by the mapped thruster forces.
- **`Test_Zero_Command`** - A zero force and torque command produces zero thruster force on every thruster.
- **`Test_Min_Shift`** - Every mapped thruster force is non negative and at least one is zero.
- **`Test_Parameter_Update`** - Applying a new thruster geometry changes the mapping used on the next tick.
- **`Test_Invalid_Parameter`** - A configuration the algorithm rejects is refused at parameter staging.

5 Appendix

5.1 Preamble

This component contains no preamble code.

5.2 Packed Types

The following section outlines any complex data types used in the component in alphabetical order. This includes packed records and packed arrays that might be used as connector types, command arguments, event parameters, etc..

Cmd_Force_Body.T:

Commanded control force in body frame components.

Table 8: Cmd_Force_Body Packed Record : 96 bits

Name	Type	Range	Size (Bits)	Start Bit	End Bit
Force_Request_Body	Packed_F32x3.T	-	96	0	95

Field Descriptions:

- **`Force_Request_Body`** - [N] Control force requested

Cmd_Torque_Body.T:

Commanded control torque in body frame components, as produced by the MRP control laws.

Table 9: Cmd_Torque_Body Packed Record : 96 bits

Name	Type	Range	Size (Bits)	Start Bit	End Bit
Torque_Request_Body	Packed_F32x3.T	-	96	0	95

Field Descriptions:

- **Torque_Request_Body** - [Nm] Control torque requested

Data_Product.T:

Generic data product packet for holding arbitrary data types

Table 10: Data_Product Packed Record : 1096 bits *(maximum)*

Name	Type	Range	Size (Bits)	Start Bit	End Bit	Variable Length
Header	Data_Product_ Header.T	-	72	0	71	-
Buffer	Data_Product_ Types.Data_ Product_ Buffer_Type	-	1024	72	1095	Header. Buffer_Length

Field Descriptions:

- **Header** - The data product header
- **Buffer** - A buffer that contains the data product type

Data_Product_Fetch.T:

A packed record which holds information for a data product request.

Table 11: Data_Product_Fetch Packed Record : 16 bits

Name	Type	Range	Size (Bits)	Start Bit	End Bit
Id	Data_Product_Types. Data_Product_Id	0 to 65535	16	0	15

Field Descriptions:

- **Id** - The data product identifier

Data_Product_Header.T:

Generic data_product packet for holding arbitrary data_product types

Table 12: Data_Product_Header Packed Record : 72 bits

Name	Type	Range	Size (Bits)	Start Bit	End Bit
Time	Sys_Time.T	-	48	0	47
Id	Data_Product_Types. Data_Product_Id	0 to 65535	16	48	63

Buffer_Length	Data_Product_ Types.Data_Product_ Buffer_Length_Type	0 to 128	8	64	71

Field Descriptions:

- **Time** - The timestamp for the data product item.
- **Id** - The data product identifier
- **Buffer_Length** - The number of bytes used in the data product buffer

Data_Product_Return.T:

This record holds data returned from a data product fetch request.

Table 13: Data_Product_Return Packed Record : 1104 bits *(maximum)*

Name	Type	Range	Size (Bits)	Start Bit	End Bit	Variable Length
The_ Status	Data_ Product_ Enums. Fetch_ Status.E	0 => Success 1 => Not_Available 2 => Id_Out_Of_Range	8	0	7	-
The_ Data_ Product	Data_ Product. T	-	1096	8	1103	-

Field Descriptions:

- **The_Status** - A status relating whether or not the data product fetch was successful or not.
- **The_Data_Product** - The data product item returned.

Desired_Control_Axes.T:

Per-axis controllability assertions for the thruster force mapping, torque xyz then force xyz, all in body frame B. A true entry asserts that axis must lie in the column space of the control mapping matrix DG.

Table 14: Desired_Control_Axes Packed Record : 48 bits

Name	Type	Range	Size (Bits)	Start Bit	End Bit
Torque_X	Boolean	0 => False 1 => True	8	0	7
Torque_Y	Boolean	0 => False 1 => True	8	8	15
Torque_Z	Boolean	0 => False 1 => True	8	16	23
Force_X	Boolean	0 => False 1 => True	8	24	31
Force_Y	Boolean	0 => False 1 => True	8	32	39
Force_Z	Boolean	0 => False 1 => True	8	40	47

Field Descriptions:

- **Torque_X** - [-] Assert the body x torque axis is controllable
- **Torque_Y** - [-] Assert the body y torque axis is controllable
- **Torque_Z** - [-] Assert the body z torque axis is controllable
- **Force_X** - [-] Assert the body x force axis is controllable
- **Force_Y** - [-] Assert the body y force axis is controllable
- **Force_Z** - [-] Assert the body z force axis is controllable

Packed_F32x24.T:

Packed 24 element vector of 32-bit floats. Holds a table of 8 three-component vectors in row major order.

Table 15: Packed_F32x24 Packed Array : 768 bits

Type	Range	Element Size (Bits)	Length	Total Size (Bits)
Short_Float	-3.40282e+38 to 3.40282e+38	32	24	768

Packed_F32x3.T:

Packed 3 element vector of 32-bit floats.

Table 16: Packed_F32x3 Packed Array : 96 bits

Type	Range	Element Size (Bits)	Length	Total Size (Bits)
Short_Float	-3.40282e+38 to 3.40282e+38	32	3	96

Packed_F32x8.T:

Packed 8 element vector of 32-bit floats. Used for thruster command arrays sized to the system thruster count.

Table 17: Packed_F32x8 Packed Array : 256 bits

Type	Range	Element Size (Bits)	Length	Total Size (Bits)
Short_Float	-3.40282e+38 to 3.40282e+38	32	8	256

Packed_U32.T:

Single component record for holding packed unsigned 32-bit value.

Table 18: Packed_U32 Packed Record : 32 bits

Name	Type	Range	Size (Bits)	Start Bit	End Bit
Value	Interfaces. Unsigned_32	0 to 4294967295	32	0	31

Field Descriptions:

- **`Value`** - The 32-bit unsigned integer.

Parameter.T:

Generic parameter packet for holding a generic parameter

Table 19: Parameter Packed Record : 792 bits *(maximum)*

Name	Type	Range	Size (Bits)	Start Bit	End Bit	Variable Length
Header	Parameter_ Header.T	-	24	0	23	-
Buffer	Parameter_ Types. Parameter_ Buffer_Type	-	768	24	791	Header.Buffer_ Length

Field Descriptions:

- **`Header`** - The parameter header
- **`Buffer`** - A buffer that contains the parameter type

Parameter_Header.T:

Generic parameter header for holding arbitrary parameters

Table 20: Parameter_Header Packed Record : 24 bits

Name	Type	Range	Size (Bits)	Start Bit	End Bit
Id	Parameter_Types. Parameter_Id	0 to 65535	16	0	15
Buffer_Length	Parameter_Types. Parameter_Buffer_ Length_Type	0 to 96	8	16	23

Field Descriptions:

- **`Id`** - The parameter identifier
- **`Buffer_Length`** - The number of bytes used in the parameter type buffer

Parameter_Update.T:

A record intended to be used as a provide/modify connector type for updating/fetching parameters.

Table 21: Parameter_Update Packed Record : 824 bits *(maximum)*

Name	Type	Range	Size (Bits)	Start Bit	End Bit	Variable Length
Table_Id	Parameter_ Types. Parameter_ Table_Id	0 to 65535	16	0	15	-

Operation	Parameter_Enums.Parameter_Operation_Type.E	0 => Stage 1 => Update 2 => Fetch 3 => Validate 4 => Discard	8	16	23	-
Status	Parameter_Enums.Parameter_Update_Status.E	0 => Success 1 => Id_Error 2 => Validation_Error 3 => Length_Error	8	24	31	-
Param	Parameter.T	-	792	32	823	-

Field Descriptions:

- **Table_Id** - The ID for the table that contains this parameter
- **Operation** - The parameter operation to perform.
- **Status** - The parameter return status.
- **Param** - The parameter that has been updated or fetched.

Sys_Time.T:

A record which holds a time stamp using EMA VTC format including seconds and subseconds since epoch (1-5-1980 to 1-6-1980 midnight).

Table 22: Sys_Time Packed Record : 48 bits

Name	Type	Range	Size (Bits)	Start Bit	End Bit
Seconds	Interfaces.Unsigned_32	0 to 4294967295	32	0	31
Subseconds	Interfaces.Unsigned_16	0 to 65535	16	32	47

Field Descriptions:

- **Seconds** - The number of seconds elapsed since epoch.
- **Subseconds** - The number of 1/(2^16) sub-seconds.

Thr_Force_Cmd.T:

Table 23: Thr_Force_Cmd Packed Record : 256 bits

Name	Type	Range	Size (Bits)	Start Bit	End Bit
Thr_Force	Packed_F32x8.T	-	256	0	255

Field Descriptions:

- **Thr_Force** - [N] Thruster force values (8 thrusters)

Thruster_Availability_X8.T:

Per-thruster availability for the thruster force mapping, one entry per thruster slot. An unavailable thruster takes no part in the mapping; a minimum of one must be available.

Table 24: Thruster_Availability_X8 Packed Array : 64 bits

Type	Range	Element Size (Bits)	Length	Total Size (Bits)
Force_Torque_Thr_Force_Mapping_Enums.Device_Availability.E	0 => Available 1 => Unavailable	8	8	64

Tick.T:

The tick datatype used for periodic scheduling. Included in this type is the Time associated with a tick and a count.

Table 25: Tick Packed Record : 80 bits

Name	Type	Range	Size (Bits)	Start Bit	End Bit
Time	Sys_Time.T	-	48	0	47
Count	Interfaces.Unsigned_32	0 to 4294967295	32	48	79

Field Descriptions:

- **Time** - The timestamp associated with the tick.
- **Count** - The cycle number of the tick.

5.3 Enumerations

The following section outlines any enumerations used in the component.

Data_Product_Enums.Fetch_Status.E:

This status denotes whether a data product fetch was successful.

Table 26: Fetch_Status Literals:

Name	Value	Description
Success	0	The data product was returned successfully.
Not_Available	1	No data product is yet available for the provided id.
Id_Out_Of_Range	2	The data product id was out of range.

Force_Torque_Thr_Force_Mapping_Enums.Device_Availability.E:

Availability of a single thruster. Values match the C DeviceAvailability_c enumeration, so Available is zero rather than the more obvious one.

Table 27: Device_Availability Literals:

Name	Value	Description

Available	0	The thruster may be used by the mapping.
Unavailable	1	The thruster is excluded from the mapping and always receives a zero force command.

Parameter_Enums.Parameter_Operation_Type.E:

This enumeration lists the different parameter operations that can be performed.

Table 28: Parameter_Operation_Type Literals:

Name	Value	Description
Stage	0	Stage the parameter.
Update	1	All parameters are staged, it is ok to update all parameters now.
Fetch	2	Fetch the parameter.
Validate	3	Validate all the parameters.
Discard	4	Discard any parameters staged by this table that have not been updated.

Parameter_Enums.Parameter_Update_Status.E:

This status enumeration provides information on the success/failure of a parameter operation.

Table 29: Parameter_Update_Status Literals:

Name	Value	Description
Success	0	Parameter was successfully staged.
Id_Error	1	Parameter id was not valid.
Validation_Error	2	Parameter values were not successfully validated.
Length_Error	3	Parameter length was not correct.